

SHETH L.U.J AND SIR M.V COLLEGE

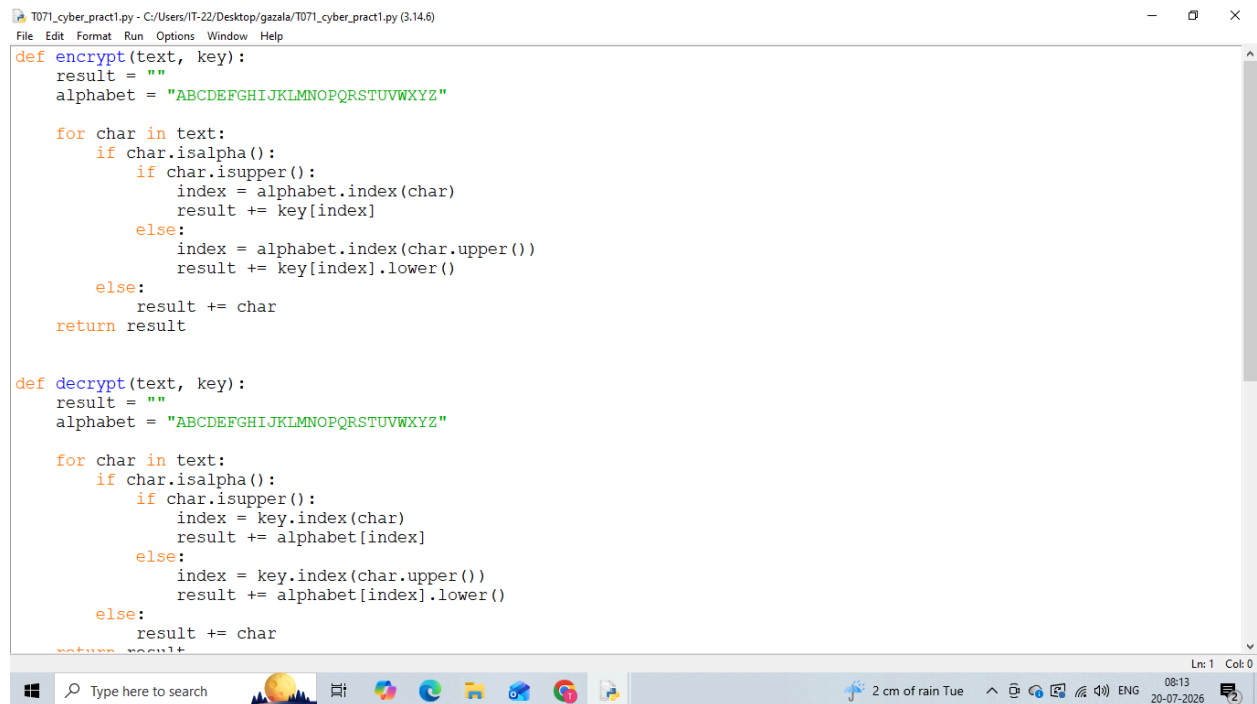
SUBJECT:- Cyber Security

Practical no: 1

Aim: 1.A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

1. B Design and implement algorithms to encrypt and decrypt messages using transposition techniques

1A:-



```
T071_cyber_pract1.py - C:/Users/IT-22/Desktop/gazala/T071_cyber_pract1.py (3,14,6)
File Edit Format Run Options Window Help

def encrypt(text, key):
    result = ""
    alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"

    for char in text:
        if char.isalpha():
            if char.isupper():
                index = alphabet.index(char)
                result += key[index]
            else:
                index = alphabet.index(char.upper())
                result += key[index].lower()
        else:
            result += char
    return result

def decrypt(text, key):
    result = ""
    alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"

    for char in text:
        if char.isalpha():
            if char.isupper():
                index = key.index(char)
                result += alphabet[index]
            else:
                index = key.index(char.upper())
                result += alphabet[index].lower()
        else:
            result += char
    return result
```

The screenshot shows a Python script in a text editor. The script defines two functions: `encrypt` and `decrypt`. Both functions take a `text` string and a `key` string as input. The `encrypt` function iterates over each character in the text. If the character is an uppercase letter, it finds its index in the alphabet and replaces it with the corresponding character from the key. If the character is a lowercase letter, it finds the index of its uppercase counterpart in the alphabet and replaces it with the corresponding lowercase character from the key. Non-alphabetic characters are added to the result unchanged. The `decrypt` function follows a similar logic, using the key to reverse the encryption process. The script is saved as `T071_cyber_pract1.py` in the directory `C:/Users/IT-22/Desktop/gazala/`. The Windows taskbar at the bottom shows the time as 08:13 on 20-07-2026, and the weather as 2 cm of rain on Tuesday.

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```
T071_cyber_pract1.py - C:/Users/IT-22/Desktop/gazala/T071_cyber_pract1.py (3.14.6)
File Edit Format Run Options Window Help

def encrypt(text, key):
    result = ""
    for i in range(len(text)):
        char = text[i]
        index = key.index(char.upper())
        result += alphabet[index].lower()
    else:
        result += char
    return result

while True:
    print("\nGazala S071 - Classical Substitution Cipher")
    print("1. Encrypt")
    print("2. Decrypt")
    print("3. Exit")

    choice = input("Enter choice: ")

    key = "QWERTYUIOPASDFGHJKLZXCVBNM"

    if choice == '1':
        text = input("Enter plaintext: ")
        print("Cipher Text:", encrypt(text, key))

    elif choice == '2':
        text = input("Enter ciphertext: ")
        print("Plain Text:", decrypt(text, key))

    elif choice == '3':
        break

    else:
        print("Invalid Choice")

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-22/Desktop/gazala/T071_cyber_pract1.py =====
Gazala S071 - Classical Substitution Cipher
1. Encrypt
2. Decrypt
3. Exit
Enter choice: 1
Enter plaintext: hello gazala
Cipher Text: itssg uqmqsq

Gazala S071 - Classical Substitution Cipher
1. Encrypt
2. Decrypt
3. Exit
Enter choice: 2
Enter ciphertext: Hello gazala
Plain Text: Pcssi oktsk

Gazala S071 - Classical Substitution Cipher
1. Encrypt
2. Decrypt
3. Exit
Enter choice: 3
>>>
```

GUI

from tkinter import *

key = "QWERTYUIOPASDFGHJKLZXCVBNM"

alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"

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```
def encrypt(text):
    result = ""
    for char in text:
        if char.isalpha():
            if char.isupper():
                result += key[alphabet.index(char)]
            else:
                result += key[alphabet.index(char.upper())].lower()
        else:
            result += char
    return result
```

```
def decrypt(text):
    result = ""
    for char in text:
        if char.isalpha():
            if char.isupper():
                result += alphabet[key.index(char)]
            else:
                result += alphabet[key.index(char.upper())].lower()
        else:
            result += char
    return result
```

```
def encrypt_text():
    text = input_text.get("1.0", END).strip()
    output_text.delete("1.0", END)
    output_text.insert(END, encrypt(text))
```

```
def decrypt_text():
```

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```
text = input_text.get("1.0", END).strip()
output_text.delete("1.0", END)
output_text.insert(END, decrypt(text))
```

```
def clear():
    input_text.delete("1.0", END)
    output_text.delete("1.0", END)
```

```
root = Tk()
root.title("Classical Substitution Cipher")
root.geometry("600x450")
root.config(bg="#EAF4FC")
```

```
Label(root,
      text="Classical Substitution Cipher",
      font=("Arial", 18, "bold"),
      bg="#EAF4FC",
      fg="#003366").pack(pady=10)
```

```
Label(root,
      text="Enter Message:",
      font=("Arial", 12, "bold"),
      bg="#EAF4FC").pack()
```

```
input_text = Text(root, height=5, width=60, font=("Arial", 11))
input_text.pack(pady=5)
```

```
frame = Frame(root, bg="#EAF4FC")
frame.pack(pady=10)
```

```
Button(frame,
      text="Encrypt",
      font=("Arial", 11, "bold"),
```

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```
bg="#4CAF50",
fg="white",
width=12,
command=encrypt_text).grid(row=0, column=0, padx=10)
```

```
Button(frame,
        text="Decrypt",
        font=("Arial", 11, "bold"),
        bg="#2196F3",
        fg="white",
        width=12,
        command=decrypt_text).grid(row=0, column=1, padx=10)
```

```
Button(frame,
        text="Clear",
        font=("Arial", 11, "bold"),
        bg="#F44336",
        fg="white",
        width=12,
        command=clear).grid(row=0, column=2, padx=10)
```

```
Label(root,
        text="Result:",
        font=("Arial", 12, "bold"),
        bg="#EAF4FC").pack()
```

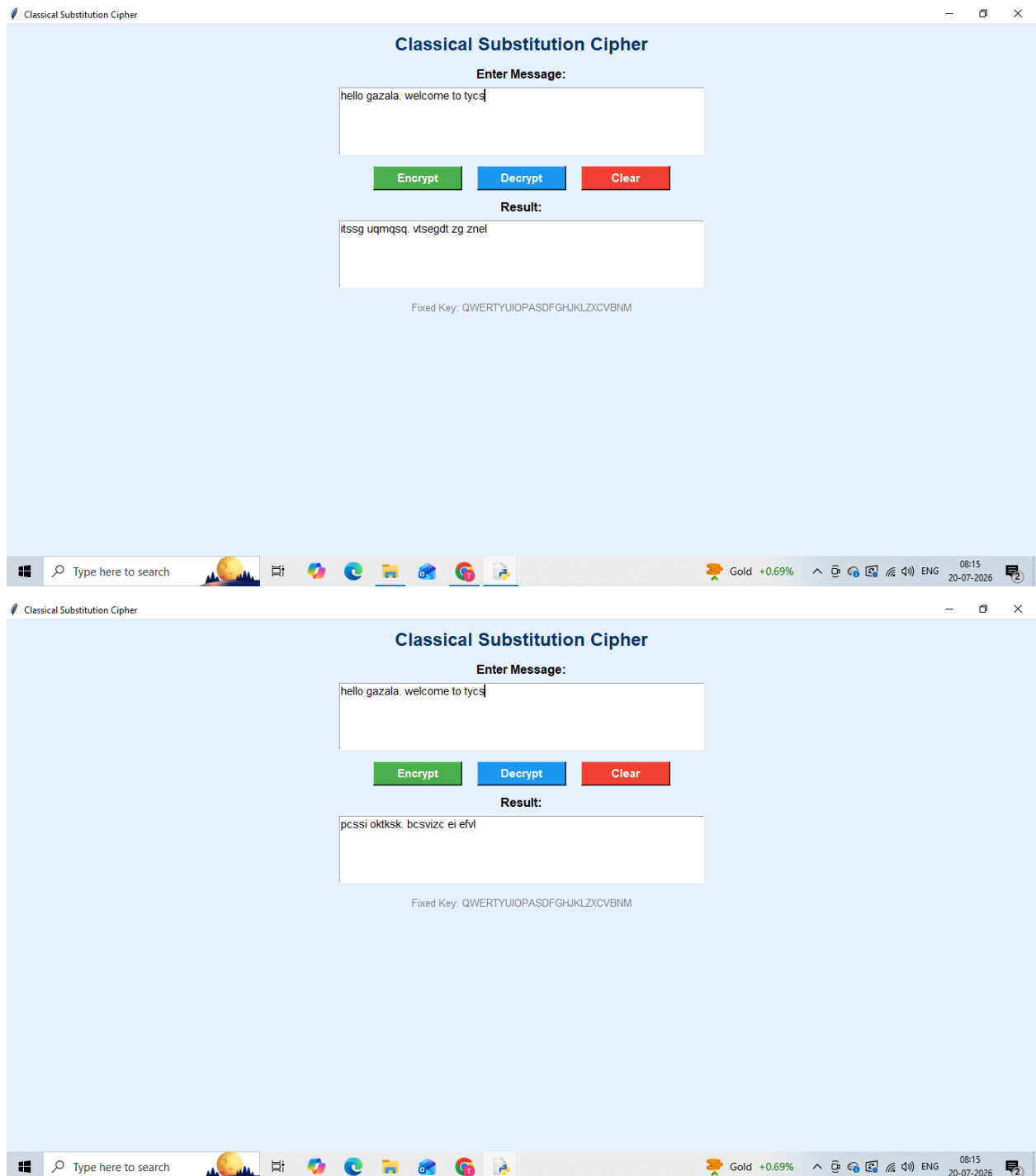
```
output_text = Text(root, height=5, width=60, font=("Arial", 11))
output_text.pack(pady=5)
```

```
Label(root,
        text="Fixed Key: QWERTYUIOPASDFGHJKLZXCVBNM",
        font=("Arial", 10),
        bg="#EAF4FC",
        fg="gray").pack(pady=10)
```

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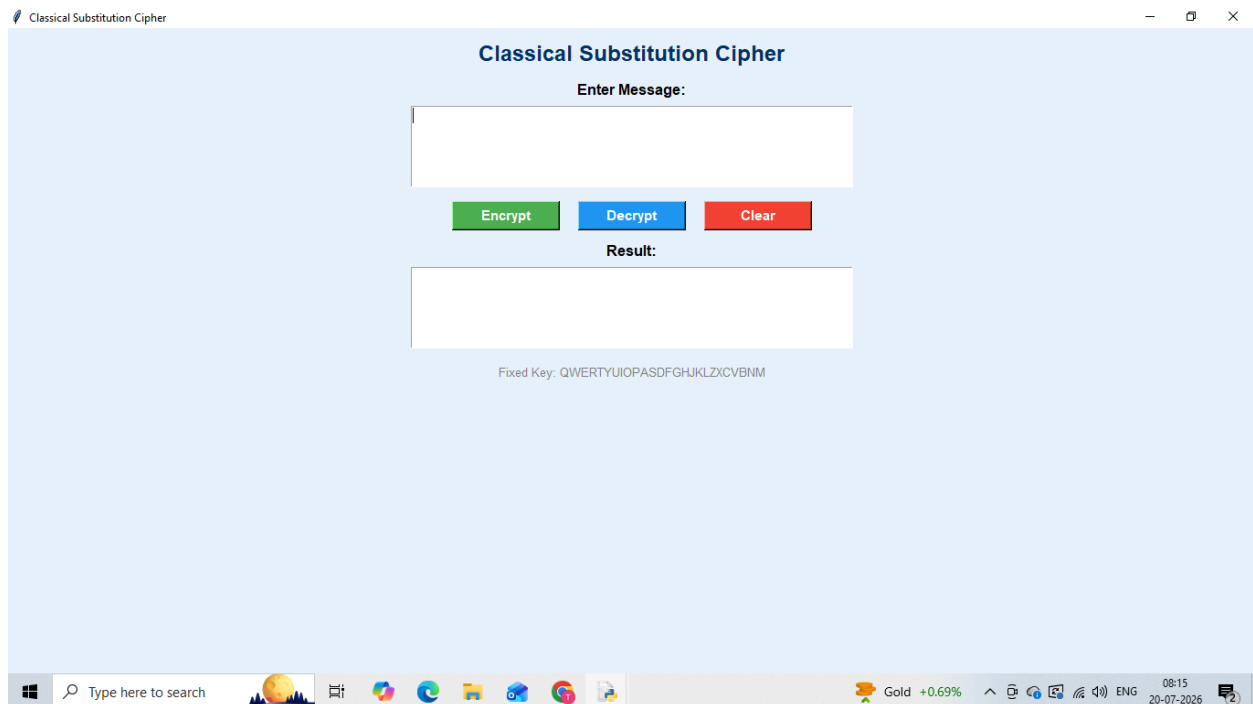
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root.mainloop()



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1B:-

```
import math
```

```
def encrypt(message, key):
```

```
    cipher = [""] * key
```

```
    for col in range(key):
```

```
        pointer = col
```

```
        while pointer < len(message):
```

```
            cipher[col] += message[pointer]
```

```
            pointer += key
```

```
    return "".join(cipher)
```

```
def decrypt(cipher, key):
```

```
    num_cols = math.ceil(len(cipher) / key)
```

```
    num_rows = key
```

```
    shaded = (num_cols * num_rows) - len(cipher)
```

```
    plain = [""] * num_cols
```

```
    col = 0
```

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```
row = 0
```

```
for symbol in cipher:
```

```
    plain[col] += symbol
```

```
    col += 1
```

```
if (col == num_cols) or (col == num_cols - 1 and row >= num_rows - shaded):
```

```
    col = 0
```

```
    row += 1
```

```
return ".join(plain)
```

```
while True:
```

```
    print("\nGazala S071 - Transposition Cipher")
```

```
    print("1. Encrypt")
```

```
    print("2. Decrypt")
```

```
    print("3. Exit")
```

```
ch = input("Enter Choice: ")
```

```
if ch == '1':
```

```
    msg = input("Enter Message: ")
```

```
    key = int(input("Enter Key: "))
```

```
    print("Cipher Text:", encrypt(msg, key))
```

```
elif ch == '2':
```

```
    msg = input("Enter Cipher Text: ")
```

```
    key = int(input("Enter Key: "))
```

```
    print("Plain Text:", decrypt(msg, key))
```

```
elif ch == '3':
```

```
    print("Thank You!")
```

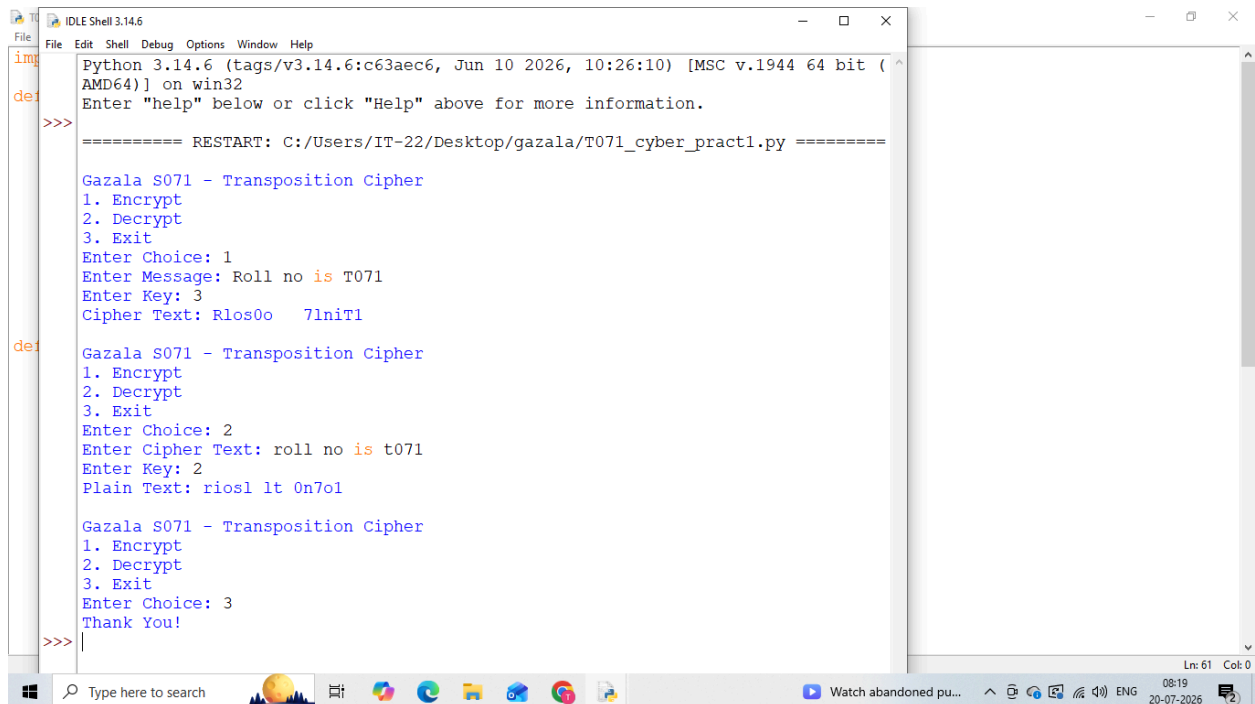
```
    break
```

```
else:
```

```
    print("Invalid Choice")
```


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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/IT-22/Desktop/gazala/T071_cyber_pract1.py =====

Gazala S071 - Transposition Cipher
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 1
Enter Message: Roll no is T071
Enter Key: 3
Cipher Text: Rlos0o 7lniT1

Gazala S071 - Transposition Cipher
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 2
Enter Cipher Text: roll no is t071
Enter Key: 2
Plain Text: riosl lt 0n7o1

Gazala S071 - Transposition Cipher
1. Encrypt
2. Decrypt
3. Exit
Enter Choice: 3
Thank You!

>>>
```

GUI:

```
import math
```

```
from tkinter import *
```

```
from tkinter import messagebox
```

```
def encrypt(message, key):
```

```
    cipher = [""] * key
```

```
    for col in range(key):
```

```
        pointer = col
```

```
        while pointer < len(message):
```

```
            cipher[col] += message[pointer]
```

```
            pointer += key
```

```
    return "".join(cipher)
```

```
def decrypt(cipher, key):
```

```
    num_cols = math.ceil(len(cipher) / key)
```

```
    num_rows = key
```

```
    shaded = (num_cols * num_rows) - len(cipher)
```

```
    plain = [""] * num_cols
```

```
    col = 0
```

```
    row = 0
```

```
    Alam Gazala T071
```

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```
for symbol in cipher:
```

```
    plain[col] += symbol
```

```
    col += 1
```

```
if (col == num_cols) or (col == num_cols - 1 and row >= num_rows - shaded):
```

```
    col = 0
```

```
    row += 1
```

```
return ".join(plain)
```

```
def encrypt_message():
```

```
    try:
```

```
        msg = txt_message.get("1.0", END).strip()
```

```
        key = int(txt_key.get())
```

```
    if msg == "":
```

```
        messagebox.showwarning("Warning", "Please enter a message.")
```

```
        return
```

```
    result = encrypt(msg, key)
```

```
    txt_result.delete("1.0", END)
```

```
    txt_result.insert(END, result)
```

```
except:
```

```
    messagebox.showerror("Error", "Enter a valid numeric key.")
```

```
def decrypt_message():
```

```
    try:
```

```
        msg = txt_message.get("1.0", END).strip()
```

```
        key = int(txt_key.get())
```

```
    if msg == "":
```

```
        messagebox.showwarning("Warning", "Please enter a message.")
```

```
        return
```

```
    result = decrypt(msg, key)
```

```
    txt_result.delete("1.0", END)
```

```
    txt_result.insert(END, result)
```

```
except:
```

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```
messagebox.showerror("Error", "Enter a valid numeric key.")
```

```
def clear_all():
```

```
    txt_message.delete("1.0", END)
```

```
    txt_result.delete("1.0", END)
```

```
    txt_key.delete(0, END)
```

```
root = Tk()
```

```
root.title("Transposition Cipher")
```

```
root.geometry("650x500")
```

```
root.configure(bg="#EAF6FF")
```

```
title = Label(root,
```

```
    text="Transposition Cipher Encryption & Decryption",
```

```
    font=("Arial", 18, "bold"),
```

```
    bg="#EAF6FF",
```

```
    fg="darkblue")
```

```
title.pack(pady=10)
```

```
Label(root,
```

```
    text="Enter Message",
```

```
    font=("Arial", 12, "bold"),
```

```
    bg="#EAF6FF").pack()
```

```
txt_message = Text(root, height=5, width=60, font=("Arial", 11))
```

```
txt_message.pack(pady=5)
```

```
Label(root,
```

```
    text="Enter Key",
```

```
    font=("Arial", 12, "bold"),
```

```
    bg="#EAF6FF").pack()
```

```
txt_key = Entry(root, font=("Arial", 12), width=10, justify="center")
```

```
txt_key.pack(pady=5)
```

```
frame = Frame(root, bg="#EAF6FF")
```

```
frame.pack(pady=15)
```

```
Button(frame,
```

```
    text="Encrypt",
```

```
    font=("Arial", 11, "bold"),
```

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```
bg="#28A745",  
fg="white",  
width=12,  
command=encrypt_message).grid(row=0, column=0, padx=10)
```

```
Button(frame,  
    text="Decrypt",  
    font=("Arial", 11, "bold"),  
    bg="#007BFF",  
    fg="white",  
    width=12,  
    command=decrypt_message).grid(row=0, column=1, padx=10)
```

```
Button(frame,  
    text="Clear",  
    font=("Arial", 11, "bold"),  
    bg="#DC3545",  
    fg="white",  
    width=12,  
    command=clear_all).grid(row=0, column=2, padx=10)
```

```
Label(root,  
    text="Result",  
    font=("Arial", 12, "bold"),  
    bg="#EAF6FF").pack()
```

```
txt_result = Text(root, height=5, width=60, font=("Arial", 11))  
txt_result.pack(pady=5)
```

```
Label(root,  
    text="Classical Cryptography - Transposition Technique",  
    font=("Arial", 10, "italic"),  
    bg="#EAF6FF",  
    fg="gray").pack(pady=15)
```

```
root.mainloop()
```

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Transposition Cipher

Transposition Cipher Encryption & Decryption

Enter Message

my name is gazala alam and my roll no is T071

Enter Key

4

Encrypt **Decrypt** **Clear**

Result

maiaaanylo 1ymasz md I T e aa r i0n gllamons7

Classical Cryptography - Transposition Technique

Windows taskbar: Type here to search, 27°C Cloudy, 08:22 20-07-2026

Transposition Cipher

Transposition Cipher Encryption & Decryption

Enter Message

my name is gazala alam and my roll no is T071

Enter Key

4

Encrypt **Decrypt** **Clear**

Result

maa yznn adonl aamim ysea lrTiao0smI7 I1g

Classical Cryptography - Transposition Technique

Windows taskbar: Type here to search, 27°C Cloudy, 08:22 20-07-2026

Alam Gazala T071

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SUBJECT:- Cyber Security

Transposition Cipher

Transposition Cipher Encryption & Decryption

Enter Message

Enter Key

Encrypt **Decrypt** **Clear**

Result

Classical Cryptography - Transposition Technique

Windows taskbar: Type here to search, 27°C Cloudy, 08:22, 20-07-2026